

1. Reduced Density operators

Alice and Bob share the quantum state

$$|\Psi\rangle = \frac{1}{\sqrt{3}} (|0\rangle|1\rangle + |1\rangle|0\rangle + |1\rangle|1\rangle),$$

where $|0\rangle$ and $|1\rangle$ are the standard orthonormal basis states for $\mathbb{C}^2$.

- Find the reduced density operator ρ_A of Alice's qubit, as a matrix in the $|0\rangle, |1\rangle$ basis.
- Consider an arbitrary density operator of a qubit σ . By expressing σ in terms of its eigenvectors and eigenvalues, show that $\sigma^2 = \sigma$ if and only if σ is a pure state.
- Compute ρ_A^2 , and hence determine whether $|\Psi\rangle$ is a product state, or entangled.
- Suppose that Alice measures the operator $Z = |0\rangle\langle 0| - |1\rangle\langle 1|$ on her qubit. Use ρ_A to find the probabilities of each possible measurement outcome, and to compute the expectation value of her measurement.
- Suppose that Bob measures the operator Z on his qubit. For each possible outcome, find the probability of obtaining that outcome, and the state of Alice's and Bob's qubits after the measurement. *Note: You will need to use $|\Psi\rangle$ (and not just ρ_A) in order to carry out this calculation.*
- Using your answer to part (e), compute the average reduced density operator of Alice's qubit after Bob's measurement. Compare this with your answer to part (a).

2. Local hidden variable models I

Alice and Bob share the entangled state

$$|\psi^-\rangle = \frac{1}{\sqrt{2}} (|0\rangle|1\rangle - |1\rangle|0\rangle).$$

and each of them measures either $X = |0\rangle\langle 1| + |1\rangle\langle 0|$ or $Z = |0\rangle\langle 0| - |1\rangle\langle 1|$ on their qubit.

- Express the state $|\psi^-\rangle$ (i) so that Alice's qubit is in terms of $|+\rangle = \frac{|0\rangle+|1\rangle}{\sqrt{2}}$ and $|-\rangle = \frac{|0\rangle-|1\rangle}{\sqrt{2}}$ (the eigenstates of X); (ii) so that Bob's qubit is in terms of $|+\rangle$ and $|-\rangle$; (iii) so that both Alice's and Bob's qubits are in terms of $|+\rangle$ and $|-\rangle$.
- Hence, or otherwise, show that the probabilities $P(a, b|x, y)$ for this experiment are

$$P(a, b|x, y) = \begin{cases} \frac{1}{2} & \text{if } a \neq b \text{ and } x = y, \\ \frac{1}{4} & \text{if } x \neq y, \\ 0 & \text{otherwise,} \end{cases}$$

where $a, b \in \{+1, -1\}$ are the measurement outcomes and $x, y \in \{0, 1\}$ are the measurement settings of Alice and Bob respectively, where $x = 0$ corresponds to measuring X , and $x = 1$ corresponds to measuring Z , and similarly for y .

- Show that the distribution $P(a, b|x, y)$ is **no-signalling**. That is, show that it satisfies

$$\sum_a P(a, b|x = 0, y) = \sum_a P(a, b|x = 1, y).$$

This condition says that Bob's marginal probabilities $P(b|x, y) = \sum_a P(a, b|x, y)$ are in fact independent of x . Explain how, if this were not true, Alice could try to send messages to Bob.

- (d) Confirm that the following local hidden variable model is able to reproduce the probabilities $P(a, b|x, y)$ for this experiment:

$P_S(\lambda, \mu)$	Alice λ		Bob μ	
	a_0	a_1	b_0	b_1
1/4	+1	+1	-1	-1
1/4	+1	-1	-1	+1
1/4	-1	+1	+1	-1
1/4	-1	-1	+1	+1